

Certificate F: Rationalization

the a priori denominator bound and exact LLL

Isaev Iskhak Khamzatovich

The «Dynamic Principle» Program — the hodge-laboratory

Standalone theorem monograph · Монография 14/16

1 Setting

Certificate F solves rationalization: recover exact rational values from numerical measurements with an a priori uniqueness guarantee. The key is the denominator bound computed before the measurements.

Theorem 1 (F1–F4). (F1) *Uniqueness: if $2\varepsilon < 1/Q^2$ with Q the denominator bound, an interval of width 2ε contains at most one rational with denominator $\leq Q$. At 100 digits the certified bound is $Q < 7.07 \cdot 10^{49}$.*

(F2) *The a priori bound is computed from geometry: $K3 - 4|\det G_8| = 256$ (Cramer), torus — 1, Klein — 2 (in the τ -basis) and 1 (over integers).*

(F3) *The integer q -scan recovers values without floating point: complexity $O(nQ)$, acceptance $d_q \leq q/2$.*

(F4) *Exact LLL: $\operatorname{Im} \tau = \sqrt{7}/2$ recovered with the minimal polynomial $4X^2 - 7$; degree-2 polynomials are unique.*

Proof. (F1) The diophantine estimate: two distinct reduced fractions with denominators $\leq Q$ differ by at least $\frac{1}{qQ} \geq \frac{1}{Q^2}$; an interval of width $< 1/Q^2$ cannot contain both. (F2) By Cramer’s rule the target coordinates are ratios of determinants built from the exact intersection matrix; denominators divide $\det G_8 = 64$; the bound $4 \cdot 64 = 256$ has margin. (F3) For each candidate $q \leq Q$: $d_q = \min_n |qx - n|$ in integer multiplications; accept at $2d_q \leq 1$. (F4) LLL on the Klein period lattice: exact operations in $\mathbb{Q}(\sqrt{-7})$ give $\operatorname{Im} \tau = \sqrt{7}/2$; the polynomial $4X^2 - 7$ is unique; $j(\tau) = -3375$ with deviation $1.99 \cdot 10^{-117}$. $\square$

Protocol data: the end-to-end K3 run

49 measurements $\rightarrow$ recovery $[Z] = [\text{target}]$ exact; maximal denominator 4; the a priori bound $Q = 256$ (margin $\times 64$); torus: degrees (2, 2); Klein: $\operatorname{Re} \tau = 1/2$, $\operatorname{Im} \tau = \sqrt{7}/2$, $\operatorname{tr} t = -1$, $\operatorname{norm} t = 2$, $j = -3375$; the $\operatorname{Im} \tau$ residual $7.75 \cdot 10^{-121}$. Negative test: the value $4\pi^2$ is rejected by the scan — consistent with Lindemann transcendence.

Remark 2 (the role of the negative test). The rejection of $4\pi^2$ is a power control: the scan does not accept “similar” values. The agreement with Lindemann’s theorem (the transcendence of π) independently confirms the correctness of the acceptance criterion.

Summary of results: summary

Certificate F accepted (5/5): rationalization with a guaranteed bound works end-to-end; the negative test passes.

Verification: reproduction

`python3 laboratory.py` $\rightarrow$ “Certificates $\rightarrow$ F”; the designer: an arbitrary value and bound Q — the lab recovers and prints the verdict.

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